Actuators for Robot

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Note: Kindly refer here (https://ajaygunalan.github.io/blog/notes/motor/motor.html) for the various definition and derivation. Here the focus is only to understand which factors need to be maximised and minimised for robots of the future.

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What is a robot ?

A robot is a bunch of actuator connected by a link(a structure). Actuators, being one of the fundamental element of a robot, will determine the core capabilities and limitation of a robot. Thus a thorough understanding of the actuator will help us to optimise it based upon our requirment. In this article we restrict ourself to Electromagnetic Actuator (Motors), since it is the most widely used one and has a higher energy efficiency compared to others like hydraulic, pneumatic, etc. Out of all the various type of motor like Induction, Reluctance, PMSM/BLDC, DC, Stepper, etc. Our goal is to understand various factors of EM actuator which influences the perfomance and hopefully to justify why pancake shaped, Quasi Direct-Drive, Permanent Magnet Synchronous Motors(PMSM), with high bandwidth 4Q control drivers and compliance will be the norm of the robotics in future. Note: (PMSM) is physically same as the BLDC Motor. We call it PMSM/BLAC when we apply a sinuosoidal emf and BLDC when we apply trapezoidal emf.

Is Actuator a deal ?

It should just spin with some toqrue, right? what else would be expected. Well, traditionaly automation and industrical robotic arms are fixed inside a cage where all the things are archietectured as per the requirment. This mean there was less uncertainity and they didn't come in contact with an unkown object. Everything was like orchestra and the problem was reduced to finding the correct set of motor commands to do a particular task satisfying some constarint (like time, cost, power, etc.). For these type of problem, yes it should spin to a postion with enough torque. Usually this was achieved by numerous trial and error and finally hard coding the values in real-time. The actuator was designed to maximise the torque by having huge gearbox like shown below.

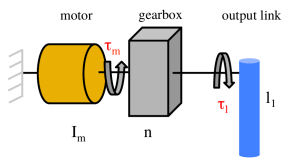

Figure 1: Acutator Design in Automation, Source[1]

Good for automation, but when comes to general purpose robotics, we except it to touch and manipulate objecsts like us and to be with us side-by-side safely, which demands acutuators to be compliant, backdriveable, energy-efficiency and force transperancy (ability to sense output torque form motor current). This can't be achieved with pure postion control and huge gearbox. If we did applied the same principles' which worked for automation, the result will like below:

Infact you can see more failures in this video of DARRPA robotics challenge where the robot failed to do task which seems trival to us. The possible reason for failure are huge gearbox, pure postion control, lack of compliance(some were rigid like bricks) and bad control algorithm(wont cover about it here). Let us delve into each of the reasons breifly.

Gearbox - The Culprit

Gearbox is an power transmitter, used to increase the torque by decreasing the speed. Addition of gearbox induces various factors such as reflected inertia, blacklash, internal stresses, coulomb, viscous friction, and stiction in the gear train. These non-linear effects hide forces acting at the output from being sensed by the motor causing motor current to be a poor indicator of output torque. Direct-Drive means directly coupling the load with actuator completely removing the gearbox. This overcomes all the drawbacks of the gearbox but increases the weight and volume of the motor due to higher torque requirment. Thus there is a trade-off between force-transperancy and reflected inrtia, torque requirment. Thus by having single-stage, backdriveable gears with low reduction ration(5-10) we achieve a decent compromise between torque requirment and force transperancy, impedance. We call this as Quasi Direct Drive.[1]

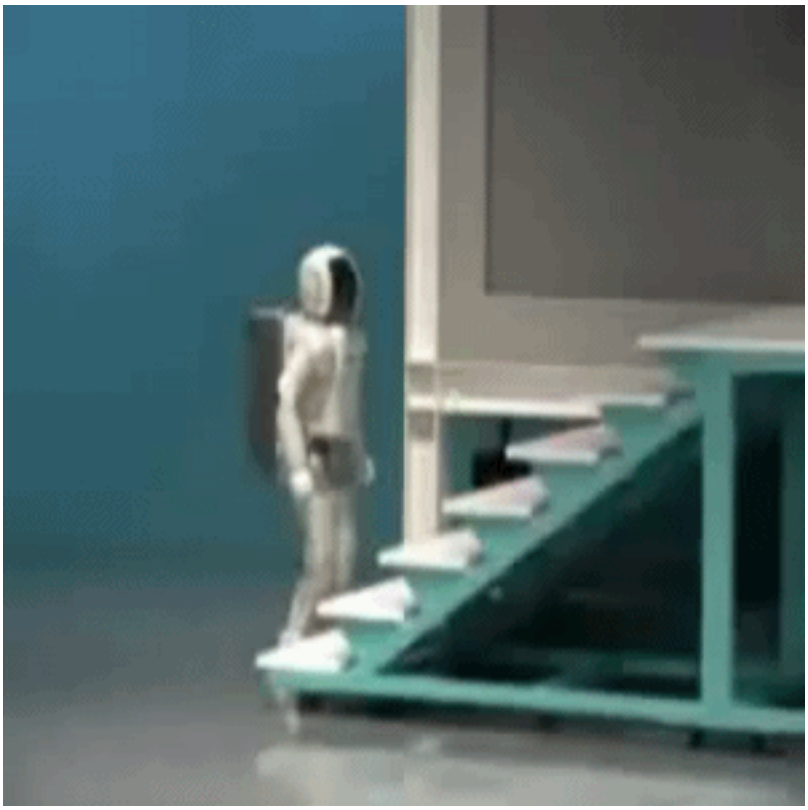

Figure 2: Result of applying automation principles in a robot

Motor control loops

If you have ever done a simple line following a robot, you would probably know that there is something called motor driver which in essence applies the required voltage based upon the user command. Based upon the driver we can give different commands such as postion, velocity, current(torque). No matter which command you give ultimatley it boils down to applying a volatge to the motor. The value of voltage will be determined by the error between desired value and actual value. This means that for a each loop (say velocity), there must be partciular sensor measuring it.

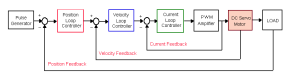

Figure 3: Current-Velocity-Position-Loops

There is reason behind why current is the inner-most loop is they are closer to the voltage(intuitively) and thus inner- loop operate (5-10) times faster than the outer loop. A motor drive can include any combination of three types of control loops—a position loop, a velocity loop, and a current. The bandwidth, or response time, of the system is a measure of how fast it responds to the changing input command. As you might have guess the inner loops have higher bandwidth. While higher bandwidth generally provides stiffer motor performance, decreases error, and improves transient response time, there are also drawbacks to high bandwidth in systems. Specifically, the higher the bandwidth, the higher the frequency at which the motor responds to disturbances, which typically requires higher accelerations and forces. Power dissipation has a squared relationship to force, so any increase in bandwidth significantly increases power dissipation (i.e. heat), and therefore, the temperature rise of the motor. Motor drivers are also classified as 1Q or 4Q drivers. 1Q allows quadrant 1, and 3, to be controlled thus only allowing the motor torque and speed to controlled in same direction. Whereas 4Q allows all the four quadrant to be controlled and thus making active electromagnetic braking, regenerating current form the motor to the battery, possible. Thus a sufficiently high bandwidth 4Q motor driver is a must.

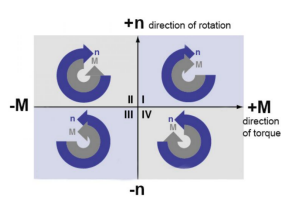

Figure 4: Torque vs Speed

Passive compliance

Compliance can be added mehcanically(using springs, damper) or virtually by control algorithm which can vary the stiffness and damping on the go. So it seems that virtual spring is better because you get a variable stiffness. Not neccessarliy, meachnical spring make control a bit complicated, but they are of enoromous use when it comes energy efficiency. They store the sudden high impact force and can be reused in next cycle. So in esscence we should select tha passive compliance for a baseline value and use software on top it for fine tuning.

Pan-Cake Shape

When we say pan-cake shaped motor, it means motors whose $r_{gap} > L_{axial}$, where r_{gap} is the radial gap between center and rotor and L_{axial} is the axial lenght of the motor. The reason is Torque density increases with r_{gap}. But it doesn't mean we can increase the r_{gap} indefinitely as Torque/inertia is inversely proportional to r_{gap}

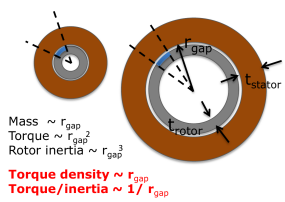

Figure 5: Gap Radius, Source[2]

Conclusion

Hopefully, I have justified why we need a high bandwidth, 4Q control driver with pan-cake shaped motor, quasi-direct-drive and compliance for the robots of the future.

![Actuators for robotics(https://ajaygunalan.github.io/projects/asset/past/motor/final.png){width=30% height=25%}

References

[1] S. Kalouche, "Design for 3d Agility and Virtual Compliance Using Proprioceptive Force Control in Dynamic Legged Robots," PhD thesis, Carnegie Mellon University, 2016.

[2] S. Seok, A. Wang, D. Otten, and S. Kim, "Actuator design for high force proprioceptive control in fast legged locomotion," in *2012 IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2012, pp. 1970–1975 [Online]. Available: http://ieeexplore.ieee.org/document/6386252/. [Accessed: 12-Sep-2018]